

CHULYM, Russian Federation

WMO: 296250

Lat: 55.100N

Lon: 80.967E

Elev: 464

StdP: 14.45

Time Zone: 7.00 (E07)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.5	-27.5	-38.7	0.6	-33.5	-32.6	0.9	-27.3	23.3	18.4	20.6	17.1	3.0	200	0.632

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	83.6	65.9	80.9	65.1	78.2	64.0	69.2	78.7	67.5	76.9	65.9	74.9	8.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.0	97.6	73.5	64.1	91.1	72.0	62.3	85.4	70.4	33.7	78.6	32.3	77.0	31.0	74.9	78.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.1	17.7	15.7	-38.7	88.4	7.3	3.1	-44.0	90.7	-48.2	92.5	-52.4	94.2	-57.7	96.5
			-38.9	72.3	7.1	2.3	-44.0	73.9	-48.1	75.3	-52.1	76.5	-57.2	78.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	34.6	-1.3	4.2	19.0	38.1	52.7	62.4	65.3	60.5	49.7	36.8	19.7	5.6	(6)
(7)		DBStd	25.64	15.39	13.04	11.52	10.42	9.81	7.45	5.79	6.65	8.28	9.18	13.23	15.00	(7)
(8)		HDD50	7118	1590	1281	962	378	89	7	0	4	106	417	908	1376	(8)
(9)		HDD65	11293	2055	1701	1427	808	393	132	69	172	464	873	1358	1841	(9)
(10)		CDD50	1479	0	0	0	20	174	378	474	328	96	9	0	0	(10)
(11)		CDD65	180	0	0	0	0	13	54	78	31	4	0	0	0	(11)
(12)		CDH74	1722	0	0	0	5	203	529	615	307	63	0	0	0	(12)
(13)		CDH80	332	0	0	0	0	45	100	122	53	12	0	0	0	(13)
(14)	Wind	WSAvg	7.3	7.4	7.6	8.0	8.0	7.9	6.3	5.6	6.0	6.5	7.3	8.1	8.4	(14)
(15)	Precipitation	PrecAvg	15.0	0.7	0.4	0.4	0.8	1.3	2.0	2.3	2.2	1.4	1.3	1.1	0.7	(15)
(16)		PrecMax	24.0	1.8	1.1	0.9	2.0	3.7	8.5	6.0	4.1	3.3	3.3	2.4	1.6	(16)
(17)		PrecMin	9.4	0.2	0.1	0.0	0.0	0.4	0.4	0.4	0.6	0.1	0.4	0.1	0.1	(17)
(18)		PrecStd	3.0	0.4	0.2	0.2	0.4	0.7	1.5	1.4	0.9	0.8	0.7	0.6	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.2	33.9	44.7	72.7	84.8	86.1	87.5	85.0	81.3	67.0	46.5	34.6	(19)
MCWB			31.6	32.0	37.4	53.1	61.5	66.0	69.2	66.7	61.8	52.1	43.3	32.4	(20)	
2%		DB	28.4	30.3	39.6	65.4	79.3	82.9	83.4	80.9	74.3	60.4	42.1	32.5	(21)	
		MCWB	27.2	28.8	34.7	51.2	59.2	65.1	67.7	66.5	58.6	49.0	38.8	31.3	(22)	
5%		DB	25.0	26.8	36.8	59.7	75.0	80.1	80.4	77.3	68.9	54.6	38.6	29.2	(23)	
		MCWB	23.7	25.2	33.5	48.4	57.2	64.3	66.8	64.2	56.3	46.5	36.1	27.9	(24)	
10%		DB	21.5	23.4	34.3	54.3	70.3	76.8	77.4	73.8	64.4	49.8	34.8	25.6	(25)	
		MCWB	20.2	21.9	31.5	45.5	55.1	62.8	65.8	62.3	54.2	44.0	32.9	24.1	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.8	32.2	38.2	55.9	64.3	69.5	72.2	70.4	63.8	53.3	44.1	33.0	(27)
MDCB			33.0	33.3	42.9	68.2	80.5	81.3	82.4	78.7	73.4	62.2	45.9	34.1	(28)	
2%		WB	27.3	29.0	35.4	52.4	61.2	67.4	70.0	67.7	60.1	50.2	39.5	31.4	(29)	
		MDCB	28.5	30.4	38.6	63.6	75.9	78.8	79.6	77.4	72.0	58.1	41.8	32.6	(30)	
5%		WB	23.8	25.4	33.7	49.0	58.6	65.7	68.6	65.6	57.4	47.2	36.1	27.9	(31)	
		MDCB	24.9	26.7	36.5	59.0	71.7	76.4	77.3	75.0	66.9	53.9	38.4	29.0	(32)	
10%		WB	20.3	22.1	31.7	45.7	56.3	64.1	66.9	63.5	54.9	44.4	33.1	24.1	(33)	
		MDCB	21.5	23.5	34.2	53.9	68.0	74.0	75.0	71.8	62.9	49.5	34.8	25.5	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	14.4	16.4	17.0	16.7	20.0	18.6	17.8	19.1	18.9	14.2	11.4	13.5	(35)
MCDBR			14.6	13.7	15.9	25.1	27.1	24.1	21.8	24.6	27.6	23.7	10.9	13.2	(36)	
MCWBR			14.3	13.2	12.6	13.9	11.4	10.1	9.4	11.6	14.0	13.7	8.9	13.2	(37)	
5% WB		MCDBR	14.2	13.4	13.6	23.9	23.7	21.1	18.9	21.3	24.3	20.8	10.5	12.9	(38)	
		MCWBR	14.1	13.0	11.4	13.9	11.3	9.8	9.2	11.2	12.8	13.0	9.0	13.0	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.290	0.322	0.374	0.410	0.407	0.407	0.436	0.412	0.368	0.351	0.321	0.291	(40)	
taud		2.014	1.977	1.911	2.082	2.207	2.277	2.194	2.241	2.318	2.232	2.062	1.961	(41)		
Ebn at Noon		188	227	242	255	265	266	254	253	250	220	173	152	(42)		
Edh at Noon		22	34	46	45	42	40	42	38	30	26	21	19	(43)		
(44)	All-Sky Solar Radiation	RadAvg	298	625	1094	1484	1726	1902	1708	1433	976	499	245	191	(44)	
(45)		RadStd	25	34	64	102	181	176	119	102	109	57	42	26	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46)
(47)	Regional (11 neighbors)	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(47)

Nomenclature: See separate page